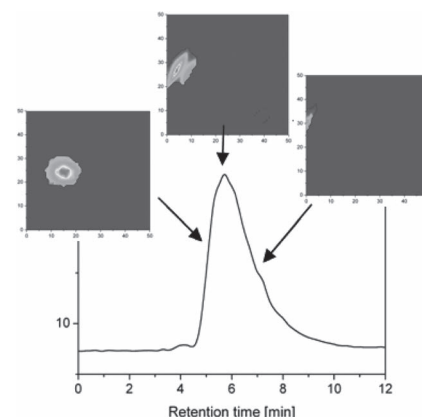


Copolymer Composition Determined by LC-MALDI-TOF MS Coupling and “MassChrom2D” Data Analysis

Steffen M. Weidner,* Jana Falkenhagen, Ingo Bressler

The MALDI-TOF MS analysis of copolymers very often results in complex spectra. A chromatographic separation/fractionation prior to the MALDI investigation can be advantageous since the MALDI mass spectra of fractions very often reveal well-resolved peaks of distinguishable copolymer series. For this purpose, different modes of chromatography have been applied. Chromatographic runs were transferred to MALDI targets utilizing a combined air/electrospray deposition device. Using the new MassChrom2D software, fraction-dependent 2D copolymer compositions plots were obtained providing additional information on the chromatographic mode, and enabling fast modification of conditions to increase separation.



1. Introduction

Polymers are typically heterogeneous with regard to molecular mass and mass distribution. Various chromatographic separation modes are known and have been increasingly applied. Size-exclusion chromatography (SEC) can be used for separation and, after calibration with suitable standards of identical structure, for mass determination. Apart from SEC, two other modes could contribute to copolymer separation as well. These are the liquid adsorption chromatography (LAC, also known as interaction chromatography IC) and the liquid chromatography at the so-called critical point of adsorption (LCCC). The principle of LCCC separation was introduced first by Entelis and co-workers.^[1–3] As

expressed in the entitled methods, these two chromatographic modes utilize (to a greater or lesser extent) adsorption effects for polymer separation. In contrast to SEC, where any adsorption interaction is ideally excluded and macromolecules are separated according to their hydrodynamic radii, interaction chromatography utilizes the interaction ability of either end groups and/or chain structure with a stationary phase. According to the chosen separation principle, this could be done in normal or reverse phase mode (NP/RP). Thus, a separation according to end groups or other chemical heterogeneities (e.g., copolymer composition,^[4] tacticity,^[5] etc.) can be achieved. A comprehensive overview about these particular techniques providing a number of examples can be obtained from Pasch and Thrathnigg.^[6]

However, the situation is even more complex for copolymers. Copolymers consist of at least two different structural units, each with its own mass distribution. Since copolymer standards are not available, a reliable calibration of the SEC cannot be achieved. Therefore, the coupling of SEC with other chromatographic methods

Dr. S. M. Weidner, Dr. J. Falkenhagen, I. Bressler
BAM—Federal Institute for Materials Research and Testing,
1.3 Structure Analysis, Richard-Willstaetter-Strasse 11, 12389
Berlin, Germany
E-mail: steffen.weidner@bam.de

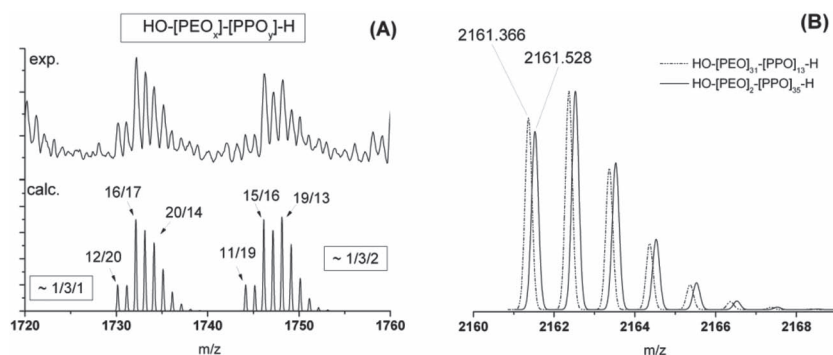


Figure 1. (A) Overlapping isotope pattern (two calculated patterns for three different copolymer compositions compared to experimental data, (B) Calculated isotope pattern for two different copolymer compositions.

and/or detectors, which are able to specifically determine the mass distribution of one copolymer component, is indispensable for a comprehensive analysis. Various examples have been reported, describing the coupling of SEC with nuclear magnetic resonance (NMR) and Fourier transform infrared (FTIR) spectroscopy.^[7–15] Other authors utilized the coupling of two chromatographic modes in a so-called two-dimensional approach.^[4,16–23] This involves the separation according to end groups in a first dimension, followed by the determination of molecular masses by SEC in the second dimension.

Since its introduction matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI) mass spectrometry (MS) have been increasingly used for polymer and copolymer analysis. However, the MALDI-TOF MS analysis of a copolymer without its previous chromatographic separation results in complex MS spectra characterized by many copolymer series overlaying each other. Even worse, in some cases, the isotope patterns of different copolymer series overlap, making a clear attribution of peaks to a distinct composition impossible. A typical example for this is represented by the class of widely used polyethylene oxide (PEO)–polypropylene oxide (PPO) copolymers. The repetition of masses of its structural units (PEO = 44.026 Da and PPO = 58.042 Da) results in isotope patterns that are composed of several overlapping series. As exemplarily shown in Figure 1a, the replacement of three PPO units (174 Da) by four PEO units (172 Da) leads to a pattern that consist of least three different series. Moreover, since the mass resolution of MALDI-TOF instruments is generally lower than those of, for example, ESI-QTOF spectrometers, even a well-resolved peak in the higher mass region of the MALDI spectrum could not easily be attributed to a certain PEO/PPO composition. As shown in Figure 1b, a rounded mass m/z 2161.5 can represent two different copolymers having completely different compositions.

Van Rooij et al.^[24] demonstrated the suitability of a MALDI-FTICR mass spectrometer for resolving overlaid isotope patterns in combination with software that was able to calculate PEO–PPO copolymer compositions from these data. However, these instruments are routinely not available and expensive. Homemade software was also used by Tortajada and co-workers.^[25] However, these authors used dihydroxybenzoic acid (DHB) and trihydroxyanthracene (dithranole) as matrix and found spectra especially differing in the low-mass region, depending on the matrix, the chosen salt, and the number

of laser shots. Moreover, a second series representing a by-product was found, which was obviously formed by a laser-induced reaction. That is clearly understandable since dithranole, and even more DHB, require much higher laser energies than DCTB.

Yet another difficulty in determining a copolymer composition could arise from their topology. As shown in a recent paper, the individual structural units of an $[A-B]_x-[A-C]_y$ copolyester could be arranged in a different way.^[26] As a result, three different polymer chains having absolutely identical masses, but different end groups, are formed. These structures are not distinguishable by mass spectrometry. Previous chromatographic separation in those cases is therefore highly desirable.

Since MALDI-TOF MS requires the embedding of samples in a suitable matrix necessary for the ionization/desorption process, its coupling with chromatographic systems still represents a challenge. The use of off-line coupling devices has been reported by several groups. These devices utilize the nebulization of the solvent by applying sheath gases at elevated temperatures^[16,27,28] and sample spotting.^[29,30] Another principle that has been applied is electro-spray deposition (ESD).^[31–37] Although the volume of sample solution that can be vaporized by ESD is limited to ca. 50 μ L, the obtained sample spots are extremely homogeneous, and the deposited amount of sample is still sufficient for MALDI. Thus, any effects resulting from an insufficient mixing of matrix, salt, and analyte were avoided. Using these off-line spray techniques, complete chromatographic runs could be transferred simply by moving the MALDI target underneath the spray device. MALDI spectra can be recorded from those target positions, which can be attributed to retention times.

The calculation of copolymer compositions was done using a new version of MassChrom2D program. In contrast to its first application for copolymers reported in 2007, a complete revision of the software was done.^[38] A new peak detection algorithm combined with new

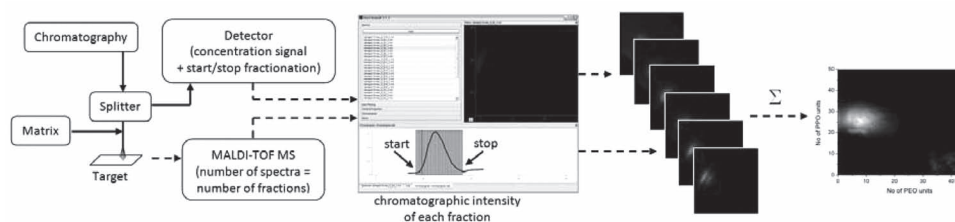


Figure 2. Scheme of measurement procedure using MassChrom2D to combine chromatography and MALDI data for determining the copolymer composition.

features to disentangle isotope patterns was implemented. Moreover, data handling and export have been completely reworked. MALDI mass spectra recorded at different positions could be easily attributed to their positions in the chromatograms. Combined with ESD of samples on the MALDI targets, a new dimension in copolymer spectra resolution can be obtained. For this work, a PEO-*b*-PPO-*b*-PEO copolymer has been exemplary chosen. These materials serve as low foaming emulsifiers in the polymerization of acrylic, styrene-acrylic, and vinyl acetate latexes, and have been widely used as low foaming nonionic surfactants, revealing excellent surface activity. Many of them are biodegradable and, therefore, environmentally friendly.

2. Experimental Section

2.1. Materials

The PEO-*b*-PPO-*b*-PEO block copolymer samples were obtained from Clariant (Sulzbach, Germany).

2.2. Liquid Adsorption Chromatography

Chromatographic experiments were carried out using a HP 1090 series HPLC system (Hewlett Packard, CA, USA) coupled to an evaporative light scattering detector (ELSD, Sedex 45, Sedere, France). A YMC RP18 column with 120 Å pore size (5 µm, 250 × 4.6 mm i.d.) was used. The system temperature was 45 °C. The eluent flow was 0.5 mL min⁻¹, and the injection volume was 20 µL. For coupling experiments, the sample concentration was 9–10 mg mL⁻¹, dissolved in a mixture of 75/25 (v/v) tetrahydrofuran (THF)/water (Roth, Germany). The critical conditions of adsorption for polypropylene oxide using a solvent system of THF/water were found at a ratio of 85/15 (v/v).^[38]

2.3. MALDI-TOF Mass Spectrometry

An Autoflex III (Bruker Daltonik, Bremen, Germany) mass spectrometer was used for the investigation. The instrument is equipped with a Smartbeam laser operating at 356 nm. Typically, 2000 laser shots were accumulated from each position. The instrument was calibrated with a series of PEG standards (Polymer Standards Service, Mainz, Germany). Spectra were recorded

in positive Reflector mode to achieve a high spectral resolution. *Trans*-2-[3-(4-*tert*-Butylphenyl)-2-methyl-2-propenylidene] malononitrile (DCTB, 10 mg mL⁻¹ in THF) was used as a matrix. Additionally, sodium trifluoroacetate (NaTFA, 2 mg mL⁻¹ in THF) was added for a selective ionization (matrix/salt, 10/1, v/v).

2.4. Electrospray Transfer Device

A homebuilt interface, consisting of a Teflon x-y table, which was adapted to the size of conventional 384 MALDI target plates, was used. A small contact enabled a contacting of the target plate to ground potential. The spray capillary (stainless steel, 0.1 mm inner diameter) was fixed in a Teflon block. The distance between target plate and capillary could be varied from 0.5 to 3 cm. High voltage (3–5 kV) was generated by a DC power supplier (FuG Elektronik GmbH, Rosenheim, Germany) and applied to the capillary. Additionally, heated gas could be applied through a series of concentric holes around the capillary to enable a better evaporation of solvents. The original solvent flow rate of 0.5 mL min⁻¹ was split to approximately 20 µL min⁻¹ by an adjustable flow splitter ASI-QuickSplit (Analytic Scientific Instruments, Richmond, CA). The matrix/salt solution (10 µL min⁻¹) was added via a t-piece to the eluent line using a micro syringe pump (Harvard Apparatus, Holliston, MA).

2.5. Software

A newly developed MassChrom2D software was used to combine chromatographic and mass spectrometric data and for calculating fraction-dependent copolymer compositions. The principal workflow is shown in Figure 2.

3. Results and Discussion

In Figure 3, the original MALDI spectra of copolymers I to IV are presented prior to fractionation. These samples reveal average molar masses between *m/z* 1500–4500. Sample I additionally shows a second peak distribution with distances of *m/z* 58 that corresponds to a PPG homopolymer possibly formed as a side product in the polymerization process. The insets of Figure 3 show the copolymer isotope patterns at masses around *m/z* 2740. They obviously consist of several individual isotope distributions overlaying each other. An attribution of such mass peaks to distinct copolymer compositions cannot be done.

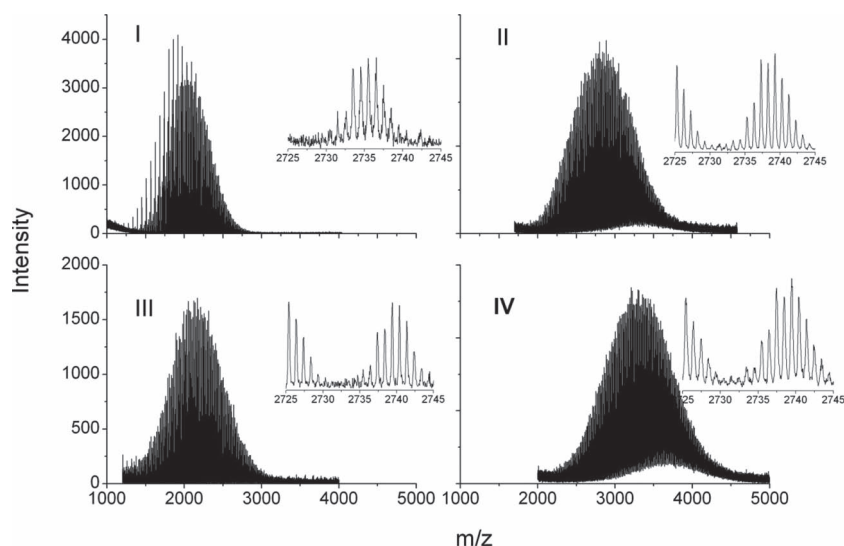


Figure 3. Original MALDI spectra of copolymers (I to IV) prior fractionation, insets show the copolymer isotope peak pattern in the region m/z 2725–2745.

In Figure 4A, the LCCC of the investigated block copolymers (I–IV) at the critical point of adsorption of PPO (dashed line at 5.2 min retention time) is shown. At these conditions, PEO units elute according to its length in SEC mode. The peak maxima can be used for a rough estimation of the size of the PEO blocks. Moreover, peaks of samples II–IV exhibit a small shoulder at retention times close to the critical point. In contrast to that, sample I elutes near the critical point and reveals, close to the peak maximum, a shoulder at lower retention times. This is in accordance with the MALDI data that indicate a considerable amount of PPO side products in sample I.

In order to increase the number of fractions that could be transferred onto the MALDI target using the transfer device, the chromatographic separation parameters were shifted away from pure critical separation towards increasing PPO adsorption. As shown in Figure 4B, the

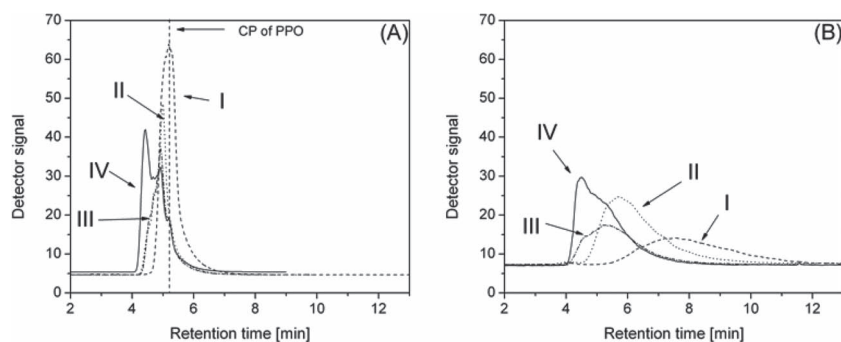


Figure 4. LAC of PEO-*b*-PPO-*b*-PEO block copolymers (I–IV) using a THF/water system (A) 85/15 (v/v) and (B) 70/30 (v/v), dashed line in (A) indicates the retention time where all PPO homopolymers with identical end groups elute at “critical” conditions of adsorption for PPO.

change of the THF/water ratio from 85/15 to 70/30 (v/v) resulted in much broader chromatograms. The resolution of the corresponding MALDI mass spectra of fractions was thereby expected to be much higher. These curves also confirm our assumption about the PEO/PPO ratio, since the observed stronger adsorption of sample I (broader peak) is caused by its higher PPO amount in the copolymer.

For the off-line fractionation experiments, two samples have been exemplarily chosen: sample I – because of its amount of PPO homopolymer, and sample IV – because this copolymer obviously contains a comparatively larger number of PEO units. Since the number of recorded MALDI spectra (34) after fractionation of sample I was too large to be shown completely, only

a representative number of spectra are presented in Figure 5. As revealed by these figures, two different chromatographic modes are observed. In the first region of the chromatogram, the copolymers eluted. MALDI mass spectra having comparatively low peak resolution were obtained from fraction 2 to 8. The average molar mass maximum of these distributions slightly decreased from m/z 2300 to 2000. This indicates a predominantly SEC-like behavior caused by the PEO units. Starting from fraction 8, in addition to copolymer distributions, well-resolved signals with peak to peak differences of 58 Da appeared. However, their molar mass maximum increased from m/z 1500 (fraction 17) to m/z 2400 (fraction 33). These peaks series also represent copolymers having comparatively low PEO amounts. Finally, in fractions 31 and 33, pure PPO homopolymer was found. The small number of peaks in these fractions indicates the presence of only a few homologous species, and, therefore, demonstrates the power of the chosen separation mode.

In contrast to this behavior, MALDI spectra of fractions of copolymer IV, depicted in Figure 6, do not clearly show the presence of PPO homopolymers. The decrease of the average molar mass maxima of these signals from m/z 3500 (fraction 2) to m/z 2000 (fraction 24) clearly indicates a SEC-like elution behavior. As seen from Figure 7A, the spectrum of one of the last fractions (no. 26 of 27) is characterized by well-resolved peak series that could be

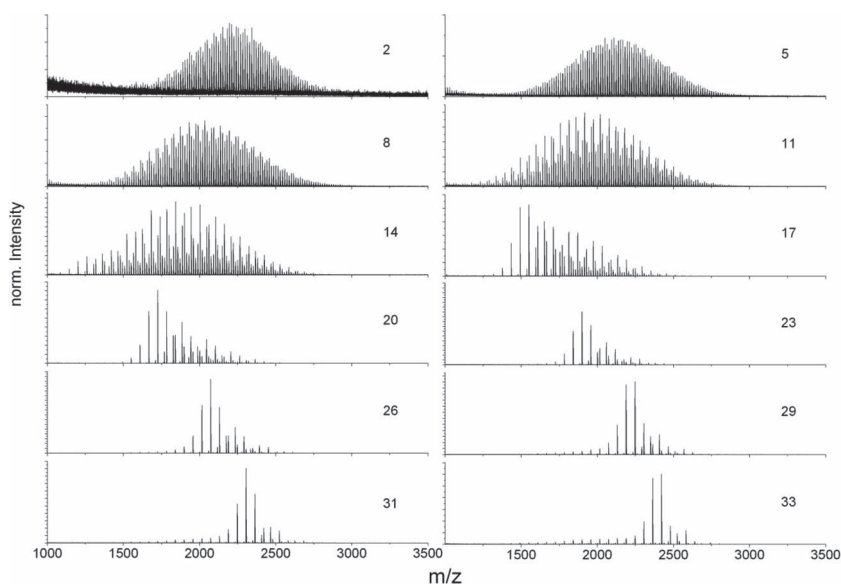


Figure 5. MALDI-TOF mass spectra of selected fractions of PEO-*b*-PPO-*b*-PEO block copolymer (I), recorded after electrospray deposition of a chromatographic run (with co-deposition of DCTB matrix).

attributed to copolymers consisting of very high numbers of PPO units, but very low numbers of PEO units, and a very short series of PPO homopolymer. In Figure 7B, a small region of the mass spectrum taken from fraction 17 (see Figure 6) is shown. Again several PEO–PPO series were found. These series consist of copolymers with a higher PEO amount. These results confirmed our initial assumption of the elution behavior deduced from chromatographic data (see Figure 4).

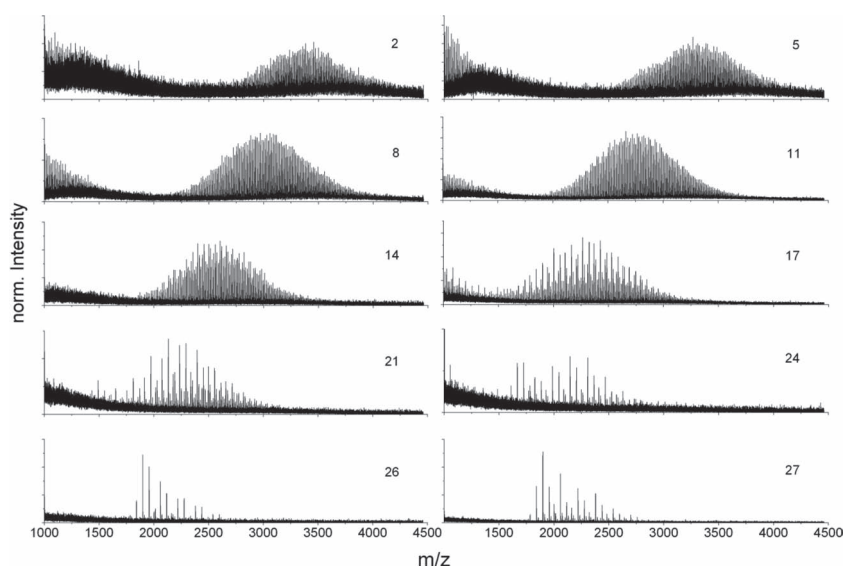


Figure 6. MALDI-TOF mass spectra of selected fractions of PEO-*b*-PPO-*b*-PEO block copolymer (IV), recorded after electrospray deposition of a chromatographic run (with co-deposition of DCTB matrix).

However, these figures also demonstrate how time consuming and complex the determination of a copolymer composition could be. Especially in the first fractions, where the peak resolution is comparatively low, a manual evaluation with regard to copolymer composition is quite impossible. For this particular reason, software has been developed that is able to handle the large amount of data obtained by an excellent chromatographic separation. Figure 8 shows the surface of the new MassChrom2D software. It basically consists of three main parts. In the upper left part, the input mask is seen. Here, molar masses of copolymer units, salt, and end groups are inserted, filter options can be determined, and ASCII data files of mass spectra series and the corresponding chromatogram can be imported. In the upper right part is a plot showing the 2D copolymer

composition. Below is the mass spectrum window. Red circles indicate those peaks that were taken for the calculation of the copolymer composition. Using filters (e.g., local maximum, local SNR) peak picking parameters can be adjusted. Baseline subtraction will be performed automatically or, if a flat baseline was not obtained, manually by defining control points. If more than one mass spectrum has been imported, this window offers the opportunity to switch between mass spectrum and chromatogram. After importing the chromatogram data file, fraction regions are automatically added, with numbers corresponding to the number of the imported mass spectra. By means of the cursor, start and end of fractionation could be determined within the chromatogram. Since MALDI intensities are not quantitative, their original intensity is normalized and combined with the intensity taken from chromatographic fractions. This approach enables a semi-quantification of MALDI data. In the 2D composition window, the composition of the copolymer of the corresponding fraction (highlighted in the chromatogram window) is shown. These data could be easily exported (e.g., csv file format) and could be post-processed in any suitable data calculation program (Excel, Origin, etc.).

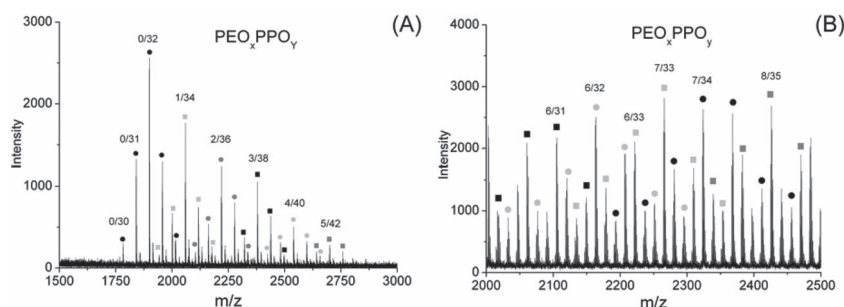


Figure 7. Detailed view of MALDI-TOF mass spectra of fraction 26 (A) and 17 (B) of PEO-*b*-PPO-*b*-PEO block copolymer (IV), recorded after electrospray deposition of a chromatographic run (with co-deposition of DCTB matrix).

The investigation started with the original copolymer I (without fractionation). Its 2D composition plot is shown in Figure 9 A. A large spot between 24 to 37 PPO units and 4 to 15 PEO units is seen. In addition, a small line at zero PEO was found indicating the PPO homopolymer having PPO numbers between 24 and 36. For a fast determination, these results might be sufficient, and could be therefore used as a fingerprint method giving a rough impression of the copolymer composition. However, this fingerprint of course is not suited to give any information on the chromatographic separation behavior.

Therefore, next the individual composition of those fractions, whose mass spectra were shown in Figure 5, was determined. As shown in Figure 10, the separation started with PEO-*b*-PPO-*b*-PEO copolymer with an average composition of 15 PEO and 25 PPO units. From fraction 2 to fraction 17, the amount of PEO constantly decreases, whereas the number of PPO units remains constant. Finally, in fraction 20 only small amounts of copolymers consisting of less than 2 PEO units were monitored. From fractions 23 to 33, PPO homopolymers with an increasing number of PPO units were found. These findings easily enable a quick overview on the chromatographic separation behavior. The separation is characterized by two

different mechanisms. As long as the macromolecules consist of PEO-PPO copolymers, a typical SEC-like elution (decreasing molar masses with increasing retention time) is found. This SEC effect is caused by a fast drop of the PEO number whereas the number of PPO units remains nearly constant. However, when the number of PEO units falls below five, this SEC effect is overlaid by additional adsorption effects caused by the PPO units. Finally, a typical separation based on an absorption mechanism (increasing retention with increasing

molar masses) was observed for PPO homopolymers.

The 2D composition plot, which was constructed by adding the 2D plots of all 34 fractions, is shown in Figure 9B. In contrast to the normalized plots shown in Figure 10, this plot utilized the intensity values delivered by the concentration detector in chromatography for every single fraction (an example for only nine fractions is shown in Figure 8—right). This information is much more reliable than any intensity value obtained by MALDI. A comparison of the both plots in Figure 9 clearly reveals some similarities, for example, the position of the copolymer spot and the existence of PPO homopolymers. However, the composed plot (Figure 9B) reveals a much higher resolution and more uniform spots showing only one maximum.

A similar comparison of both the original and combined 2D plot for another copolymer (sample IV) having higher molar masses and different composition is presented in Figure 11. From these two plots, it becomes obvious that copolymer IV has a much higher amount of PEO units than were found for copolymer I.

Again, the composite plot constructed using 27 single fractions, whose MALDI spectra were attributed to the chromatographic signal intensity (after normalization),

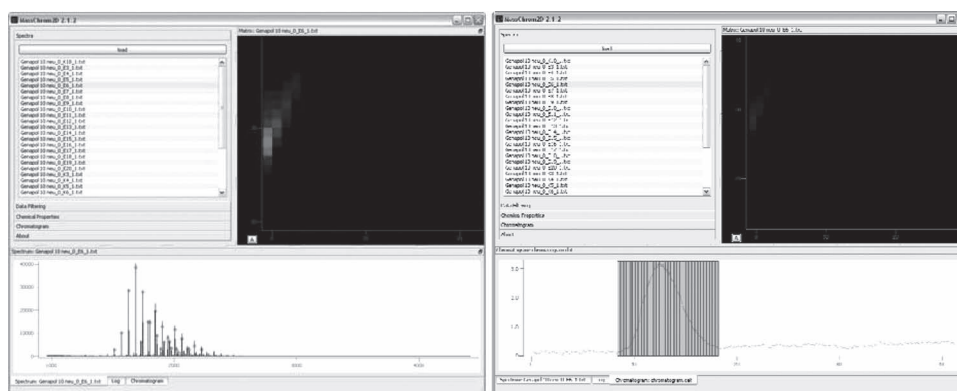


Figure 8. Appearance of MassChrom2D software for one fraction (spectrum view – left) and for fraction 6 (highlighted by a dark-blue rectangle in the bottom window) of a series of 34 fractions taken from a chromatographic run (chromatogram view – right).

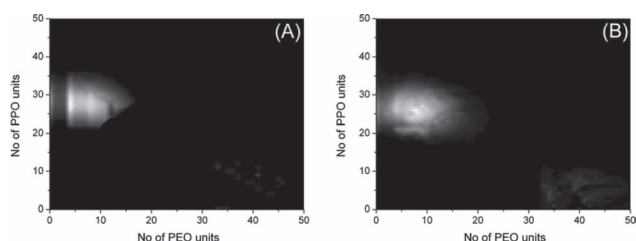


Figure 9. 2D composition plots of (A) nonfractionated PEO-*b*-PPO-*b*-PEO copolymer (I), and (B), after reconstruction using intensity values from chromatography for each single fraction, obtained by MassChrom2D software.

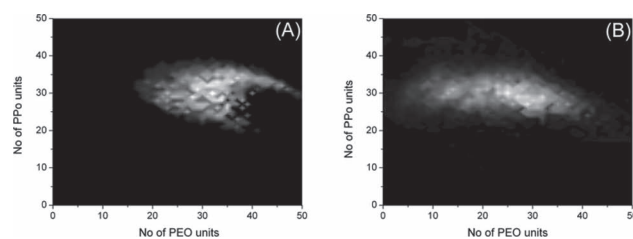


Figure 11. 2D composition plots of (A) nonfractionated PEO-*b*-PPO-*b*-PEO copolymer (IV), and (B), after reconstruction using intensity values from chromatography for each single fraction, obtained by MassChrom2D software.

shows much higher resolution and less artifacts than that of the original (nonfractionated) sample (Figure 11A).

4. Conclusion

In this work, new MassChrom2D software for the determination of copolymer composition was presented. Our examples demonstrated that this particular software can be used for fast estimation of sample composition of nonfractionated copolymer samples. However, it reveals its full potential only after fractionation, taking advantage of the power of an additional chromatographic separation. Moreover, different chromatographic separation modes for each copolymer unit were monitored. For the chosen chromatographic conditions, a typical SEC-like elution of PEO-PPO copolymers was found. However, this effect is nearly exclusively based on a fast drop of the PEO number, whereas the number of PPO units remains nearly constant. At a certain (low) number of PEO units, the SEC principle is overlaid by additional adsorption effects caused by the PPO units. Finally, a typical separation based on an absorption mechanism was observed for PPO homopolymer side product. These results enable a quick adjustment of chromatographic parameters (e.g., solvent composition and/or columns) to achieve a better separation, and/or could be used for the faster determination of “critical” adsorption conditions of one copolymer unit.

Acknowledgements: The authors would like to thank the Federal Institute for Materials Research and Testing (BAM) for technical and financial support.

Received: April 2, 2012; Revised: May 29, 2012; Published online: July 12, 2012; DOI: 10.1002/macp.201200169

Keywords: copolymer composition; chromatography; coupling; MALDI-TOF mass spectrometry

- [1] V. V. Evreinov, A. V. Gorshkov, T. N. Prudskova, V. V. Guryanova, A. V. Pavlov, A. Y. Malkin, S. G. Entelis, *Polym.Bull.* **1985**, *14*, 131.
- [2] A. V. Gorshkov, V. V. Evreinov, S. G. Entelis, *Doklady Akademii Nauk Sssr* **1983**, *272*, 632.

Figure 10. 2D composition plots of fractionated PEO-*b*-PPO-*b*-PEO copolymer (I) obtained by MassChrom2D software (normalized intensity plots).

- [3] S. G. Entelis, V. V. Evreinov, A. V. Gorshkov, *Adv. Polym. Sci.* **1986**, 76, 129.
- [4] J. Falkenhagen, H. Much, W. Stauf, A. H. E. Muller, *Macromolecules* **2000**, 33, 3687.
- [5] D. Berek, M. Janco, K. Hatada, T. Kitayama, N. Fujimoto, *Polym. J.* **1997**, 29, 1029.
- [6] H. Pasch, B. Thrathnigg, *HPLC of Polymers*, Springer Laboratory, Berlin Heidelberg New York **1998**.
- [7] G. Adamus, P. Rizzarelli, M. S. Montaudo, M. Kowalczyk, G. Montaudo, *Rapid Commun. Mass Spectrom.* **2006**, 20, 804.
- [8] E. de Goede, P. Mallon, H. Pasch, *Macromol. Mater. Eng.* **2010**, 295, 366.
- [9] N. Fandrich, J. Falkenhagen, S. M. Weidner, D. Pfeifer, B. Staal, A. F. Thunemann, A. Laschewsky, *Macromol. Chem. Phys.* **2010**, 211, 869.
- [10] I. Kramer, W. Hiller, H. Pasch, *Macromol. Chem. Phys.* **2000**, 201, 1662.
- [11] T. Macko, U. Schulze, R. Brull, A. Albrecht, H. Pasch, T. Fonagy, L. Haussler, B. Ivan, *Macromol. Chem. Phys.* **2008**, 209, 404.
- [12] M. Montaudo, *Abstracts Papers Am. Chem. Soc.* **2000**, 219, U392.
- [13] H. Pasch, L. C. Heinz, T. Macko, W. Hiller, *Pure Appl. Chem.* **2008**, 80, 1747.
- [14] I. G. Romero, H. Pasch, *E-Polym.* **2005**.
- [15] G. WilczekVera, P. O. Danis, A. Eisenberg, *Macromolecules* **1996**, 29, 4036.
- [16] J. Adrian, E. Esser, G. Hellmann, H. Pasch, *Polymer* **2000**, 41, 2439.
- [17] D. Braun, I. Kramer, H. Pasch, S. Mori, *Macromol. Chem. Phys.* **1999**, 200, 949.
- [18] G. L. Cheng, J. Cullen, C. S. Wu, *J. Chromatogr. A* **2011**, 1218, 237.
- [19] P. Kilz, R. P. Kruger, H. Much, G. Schulz, *Chromatographic Characterization Polym.* **1995**, 247, 223.
- [20] M. I. Malik, P. Sinha, G. M. Bayley, P. E. Mallon, H. Pasch, *Macromol. Chem. Phys.* **2011**, 212, 1221.
- [21] J. A. Raust, L. Houillot, M. Save, B. Charleux, C. Moire, C. Farcet, H. Pasch, *Macromolecules* **2010**, 43, 8755.
- [22] C. Schmid, S. Weidner, J. Falkenhagen, C. Barner-Kowollik, *Macromolecules* **2012**, 45, 87.
- [23] A. Siewing, B. Lahn, D. Braun, H. Pasch, *J. Polym. Sci. Part A-Polym. Chem.* **2003**, 41, 3143.
- [24] G. J. van Rooij, M. C. Duursma, C. G. de Koster, R. M. A. Heeren, J. J. Boon, P. J. W. Schuyf, E. R. E. van der Hage, *Anal. Chem.* **1998**, 70, 843.
- [25] P. Terrier, W. Buchmann, G. Cheguillaume, B. Desmazieres, J. Tortajada, *Anal. Chem.* **2005**, 77, 3292.
- [26] S. M. Weidner, J. Falkenhagen, K. Knop, A. Thunemann, *Rapid Commun. Mass Spectrom.* **2009**, 23, 2768.
- [27] L. Coulier, E. R. Kaal, T. Hankemeier, *J. Chromatogr. A* **2005**, 1070, 79.
- [28] C. E. Kassis, J. M. DeSimone, R. W. Linton, E. E. Remsen, G. W. Lange, R. M. Friedman, *Rapid Commun. Mass Spectrom.* **1997**, 11, 1134.
- [29] T. I. Stevenson, J. A. Loo, *Lc Gc-Mag. Separation Sci.* **1998**, 16, 54.
- [30] M. W. F. Nielen, *Anal. Chem.* **1998**, 70, 1563.
- [31] Y. X. Wang, Y. Zhou, B. M. Balgley, J. W. Cooper, C. S. Lee, D. L. DeVoe, *Electrophoresis* **2005**, 26, 3631.
- [32] A. S. Kononikhin, E. N. Nikolaev, V. Frankevich, R. Zenobi, *Eur. J. Mass Spectrom.* **2005**, 11, 257.
- [33] S. J. Wetzel, C. M. Guttman, K. M. Flynn, *Rapid Commun. Mass Spectrom.* **2004**, 18, 1139.
- [34] H. Wei, K. Nolkranz, D. H. Powell, J. H. Woods, M. C. Ko, R. T. Kennedy, *Rapid Commun. Mass Spectrom.* **2004**, 18, 1193.
- [35] S. D. Hanton, I. Z. Hyder, J. R. Stets, K. G. Owens, W. R. Blair, C. M. Guttman, A. A. Giuseppetti, *J. Am. Soc. Mass Spectrom.* **2004**, 15, 168.
- [36] X. W. Lou, J. L. J. van Dongen, *J. Mass Spectrom.* **2000**, 35, 1308.
- [37] R. R. Hensel, R. C. King, K. G. Owens, *Rapid Commun. Mass Spectrom.* **1997**, 11, 1785.
- [38] S. M. Weidner, J. Falkenhagen, S. Maltsev, V. Sauerland, M. Rinken, *Rapid Commun. Mass Spectrom.* **2007**, 21, 2750.